

SPECIFICATION — NODE 2 — PAGE 4

INTRAVAGINAL LONGITUDINAL WEARABLE — NODE-SPECIFIC EXEMPLAR CLAIM-LIKE STATEMENTS (OPTIONAL, NON-LIMITING)

A. Overview

This page provides optional, non-limiting claim-like statements for Node 2 to support standalone protectability of the intravaginal longitudinal wearable. A provisional application does not require claims. These statements are illustrative only and do not limit the disclosure.

B. Exemplary Node 2 System Statements (Non-Limiting)

1. A wearable device comprising: a) a sealed body configured for intravaginal placement during a prolonged wear session; b) at least one sensing modality selected from bio-impedance sensing, capacitive wetness/dielectric sensing, pH sensing, temperature sensing, pressure sensing, and inertial sensing; and c) a communications interface configured to provide time-series data to a computing system; wherein the computing system computes at least one quality index and outputs privacy-preserving longitudinal metrics. 2. The wearable device of statement 1, wherein bio-impedance sensing comprises multi-frequency impedance sensing using one or more electrode pairs. 3. The wearable device of statement 1, wherein the device includes a capacitive wetness/dielectric sensing region configured to generate a hydration/moisture trend index. 4. The wearable device of statement 1, wherein pH sensing comprises solid-state pH sensing including an ISFET-based pH sensor. 5. The wearable device of statement 1, wherein the device includes distributed pressure sensing configured to generate a pelvic activity index. 6. The wearable device of statement 1, wherein the inertial sensing comprises a 6-axis IMU configured for motion gating and artifact rejection. 7. The wearable device of statement 1, wherein the computing system defines valid windows by applying threshold criteria to at least two quality indices comprising motion gating and contact integrity. 8. The wearable device of statement 1, wherein the computing system computes an electrode integrity index based on impedance plausibility and signal continuity. 9. The wearable device of statement 1, wherein privacy-preserving longitudinal metrics comprise one or more of moisture/hydration indices, impedance-derived tissue state indices, pH trend indices, pelvic activity indices, thermal context tags, stability indices, and confidence indicators.

C. Exemplary Node 2 Method Statements (Non-Limiting)

10. A method comprising: a) acquiring time-series signals from an intravaginal wearable during a prolonged wear session; b) computing at least one quality index including motion gating and contact integrity; c) selecting valid windows based on the quality index; and d) outputting privacy-preserving longitudinal metrics comprising indices and confidence indicators. 11. The method of statement 10, further comprising computing a multi-frequency impedance-derived tissue state index from selected valid windows. 12. The method of statement 10, further comprising computing a moisture trend index using capacitive wetness/dielectric sensing. 13. The method of statement 10, further comprising computing a pH trend index and gating the pH trend index using a pH sensor quality index. 14. The method of statement 10, further comprising computing a pelvic activity index from distributed pressure sensing and rejecting windows based on IMU-informed motion gating.

D. Optional Integration Statements (Non-Limiting)

15. The system or method of any prior statement, further comprising time-aligning derived metrics from the intravaginal wearable with derived metrics from an upper-body context node to select matched context windows. 16. The system or method of any prior statement, further comprising correlating derived metrics from the intravaginal wearable with derived metrics from at least one additional node to compute an alignment score and a confidence indicator.

E. Variations

Any statement herein may be modified to include additional sensors, fewer sensors, substituted equivalent modalities, different electrode layouts, different housing geometries, different duty cycles, and different processing distributions (on-node, on-device, cloud) while maintaining the functional intent of intravaginal longitudinal capture, validity gating, and privacy-preserving outputs.